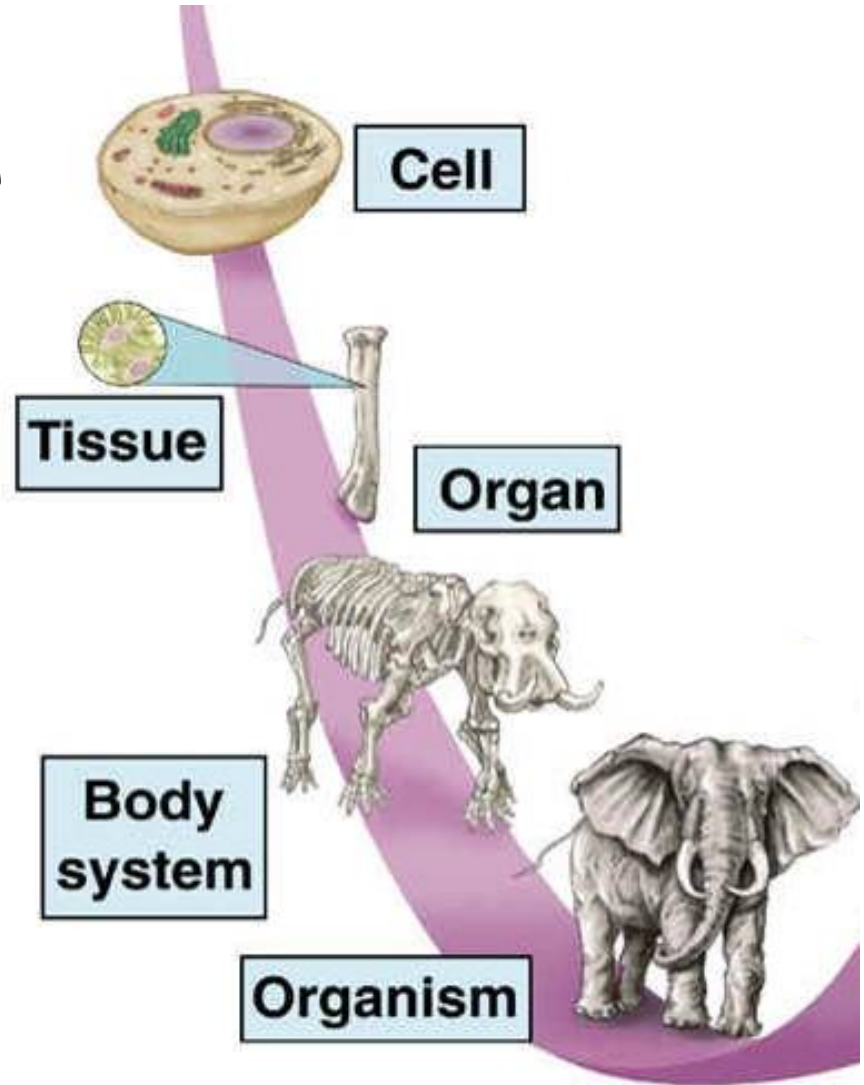
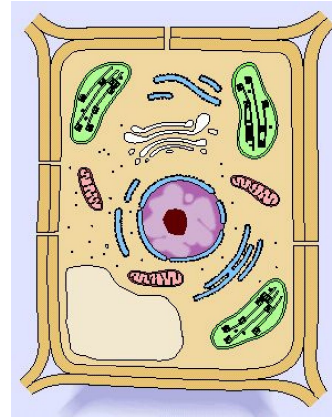


Levels of Organization Notes

TEKS 7.12C
recognize levels of
organization in
plants and animals,
including cells,
tissues, organs,
organ systems, and
organism



Cell: Basic unit of structure & function in all living organisms.



Examples:

animals- muscle, bone, nerve, blood, skin, egg, sperm, fat

plants- epidermal, parenchyma, spongy, cambium, xylem, phloem

Tissue: Similar cells working together to perform a specific function.



Examples:

animals- muscle, nervous, epithelial,
connective

plants- dermal, ground, vascular

Organ: Different tissues
working together to
perform a specific
function.



Examples:

animals- brain, stomach, intestines,
heart, lungs, liver, etc...

plants- leaves, buds, flowers, fruits,
stems, roots, tubers, rhizomes

Organ system: Group of different organs working together to perform a particular set of functions.



Examples:

**animals- circulatory, respiratory,
nervous, digestive, reproductive,
muscular, skeletal, etc...**

plants- shoot, root

**Organism: Anything
living that is made of
cells & needs energy,
excretes wastes,
grows, responds,
reproduces, & uses
gases.**



**Examples: animals, plants, bacteria,
fungi, protists, monerans**

Population

- Group of the same species living in a given area.
- Example: the students in this classroom are a population

Community

- Groups of different populations living in a given area.
- Example: Your neighborhood (all the people and pets that live around you)

Ecosystem

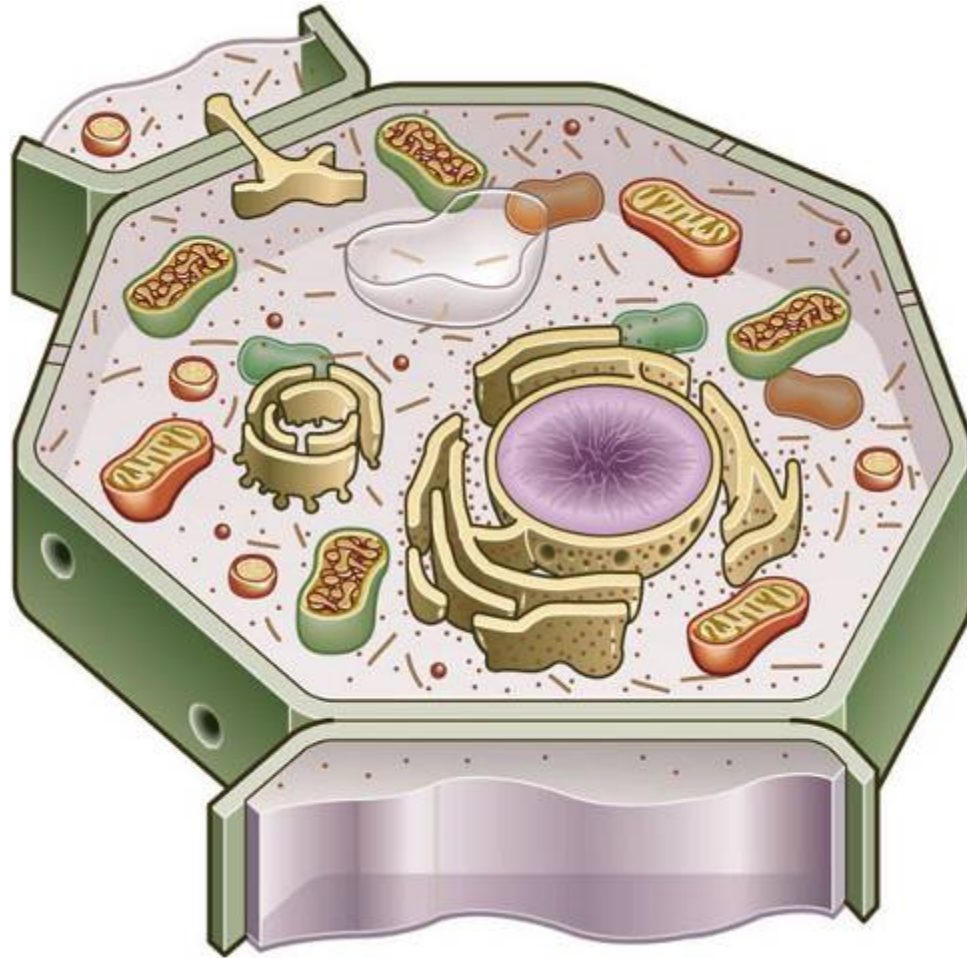
- All the living and non living things found in a given area.
- Example: All the trees, animals, rocks, water, dirt, etc... found in a park.

What level is it? **animal
organism**



<http://spyglasshill.files.wordpress.com/2009/02/red-eye-tree-frog-7550-13.jp>

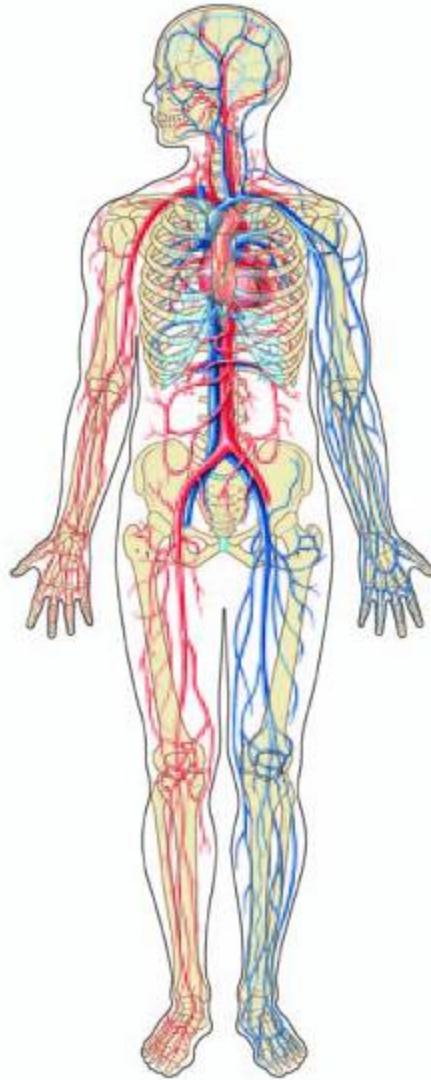
What level is it? plant cell



<http://fds.oup.com/www.oup.co.uk/images/oxed/children/yoer/nature/plantcell.jpg>

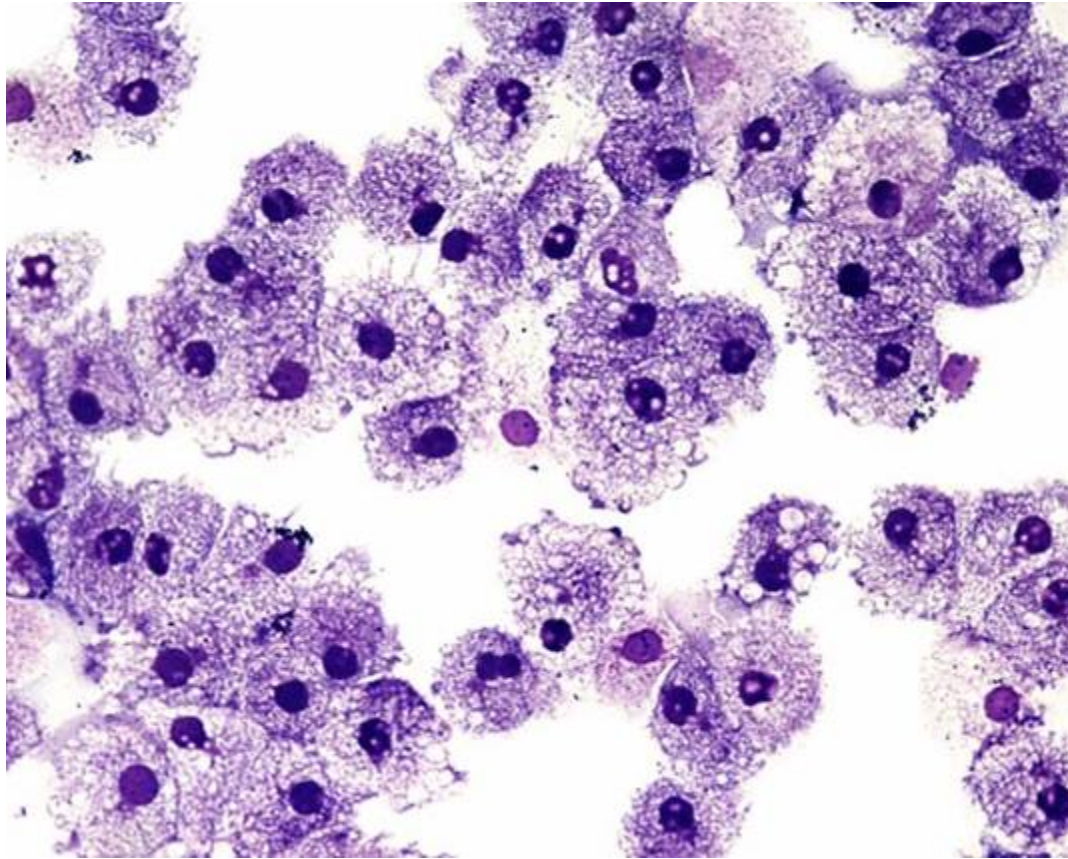
What level is it?

animal system



What level is it?

animal
tissue



http://www.fi.edu/learn/heart/blood/images/large_white-cells.jpg
g

What level is it?

plant
organ



[http://naturallyadvanced.files.wordpress.com/2009/11/bamboo.jp](http://naturallyadvanced.files.wordpress.com/2009/11/bamboo.jpg)

What level is it?

animal
organism

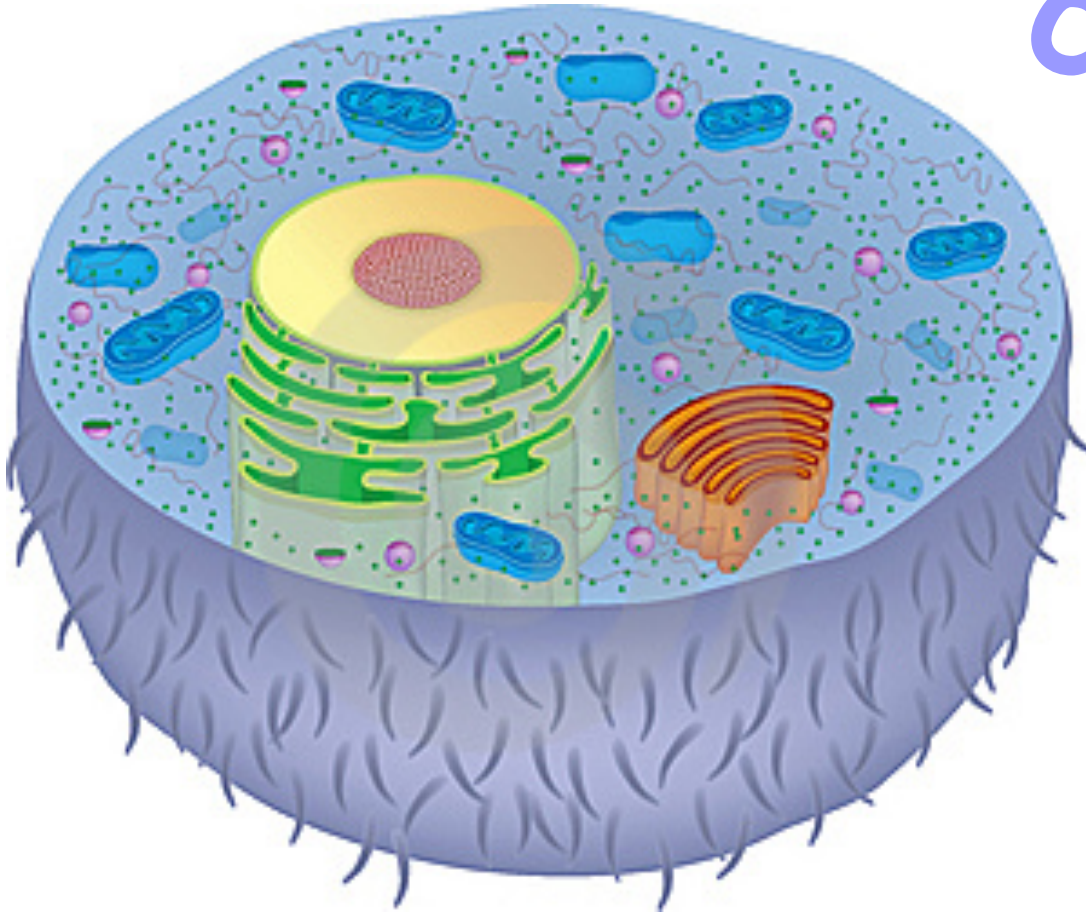


<http://pixdaus.com/pics/1264474052QxHflmc.jp>

g

What level is it?

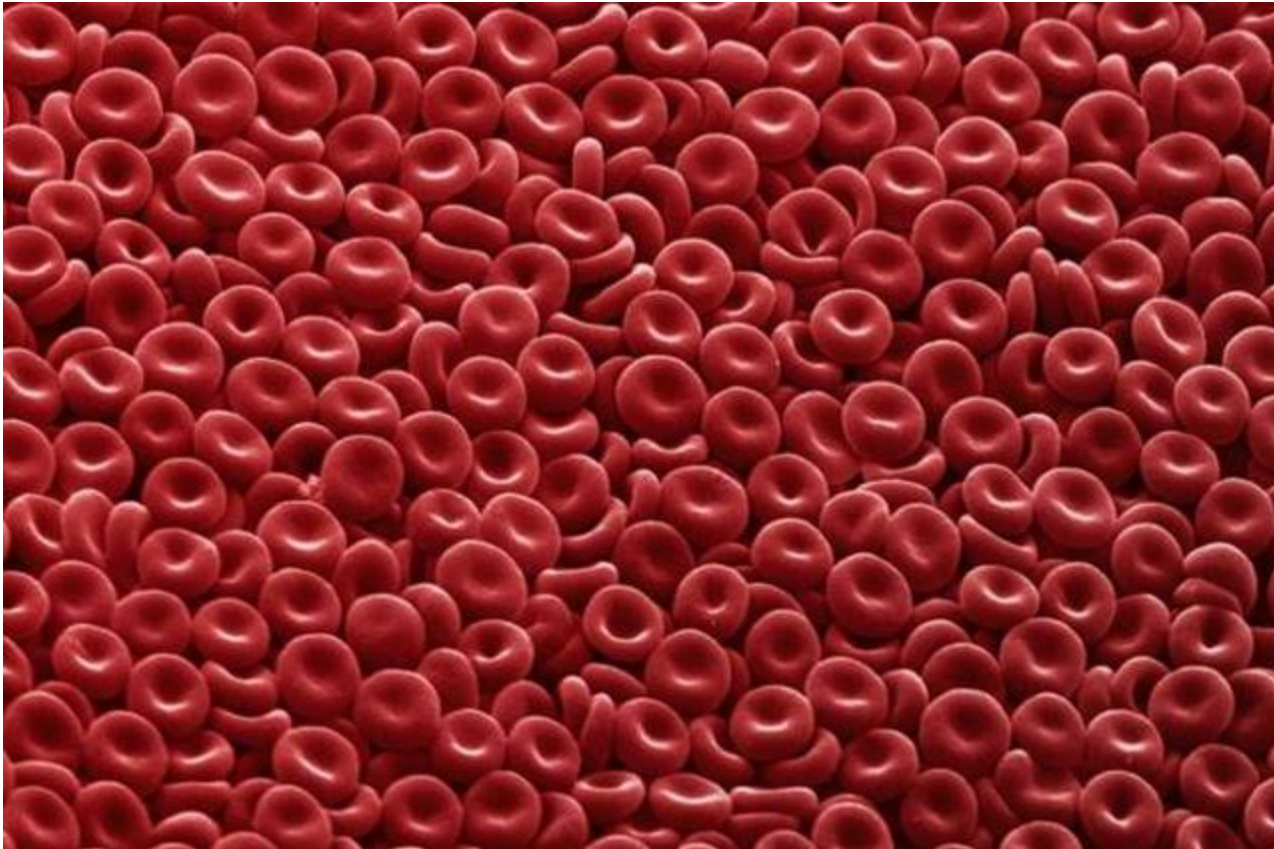
animal cell



http://thumbs.dreamstime.com/thumb_332/1226501549fUKXGc.jp

g

What level is it? **animal
tissue**



[http://gooddeedaday.files.wordpress.com/2009/04/red-blood-cells.jp
g](http://gooddeedaday.files.wordpress.com/2009/04/red-blood-cells.jpg)

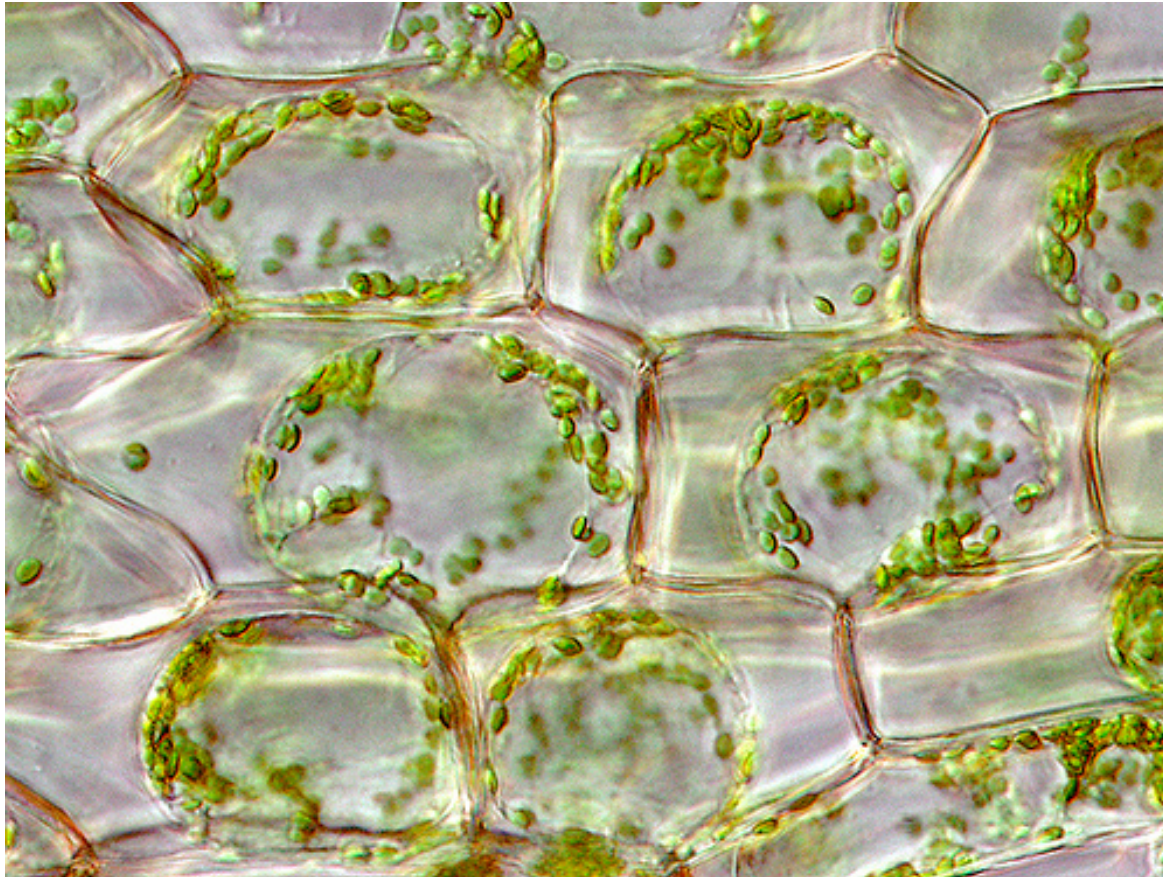
What level is it? **animal
organism**



[http://ic2.pbase.com/v3/14/60114/1/49063498.MacleaysSpectre_Extatosomatiaratum.jp](http://ic2.pbase.com/v3/14/60114/1/49063498.MacleaysSpectre_Extatosomatiaratum.jpg)

What level is it?

plant
tissue



[http://farm3.static.flickr.com/2306/3004518176_fa96cf3033.jp](http://farm3.static.flickr.com/2306/3004518176_fa96cf3033.jpg)

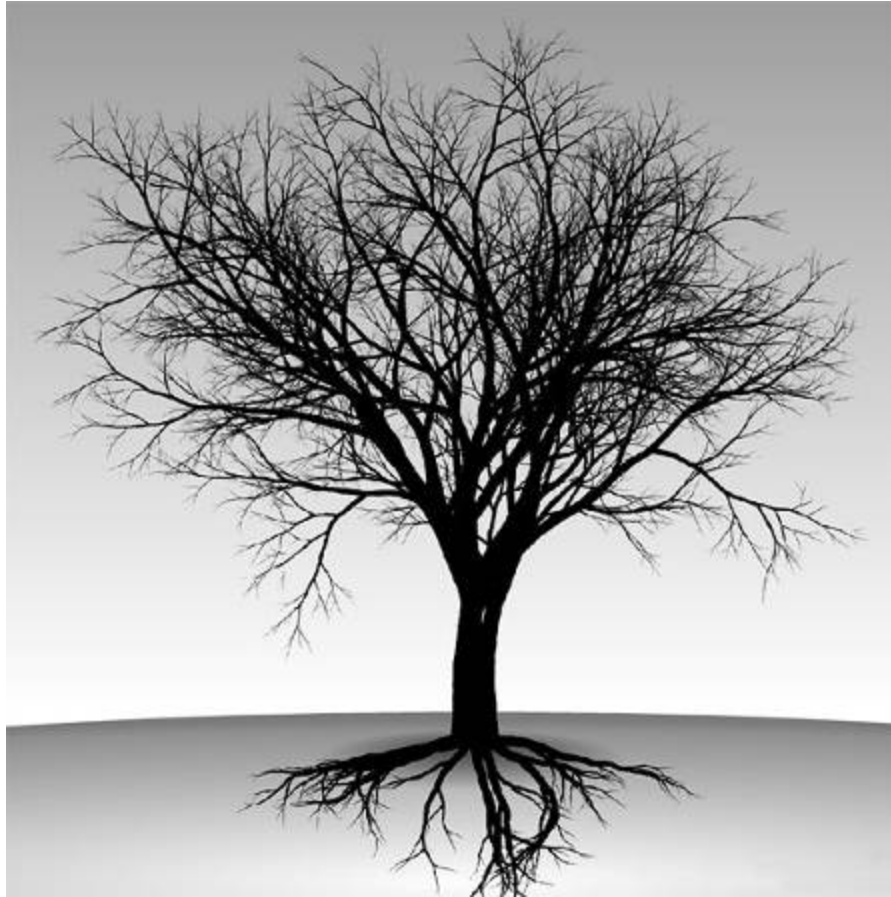
What level is it?

animal
system



[http://en.wikivisual.com/images/6/6f/Swordfish_skeleton.jp](http://en.wikivisual.com/images/6/6f/Swordfish_skeleton.jpg)

What level is it? plant
organism



<http://billylai.files.wordpress.com/2009/05/elm.jpg>

g

What level is it?

**plant
organ**



[http://www.kevinwolf.com/images/Autumn%20Leaf.jp](http://www.kevinwolf.com/images/Autumn%20Leaf.jpg)

g

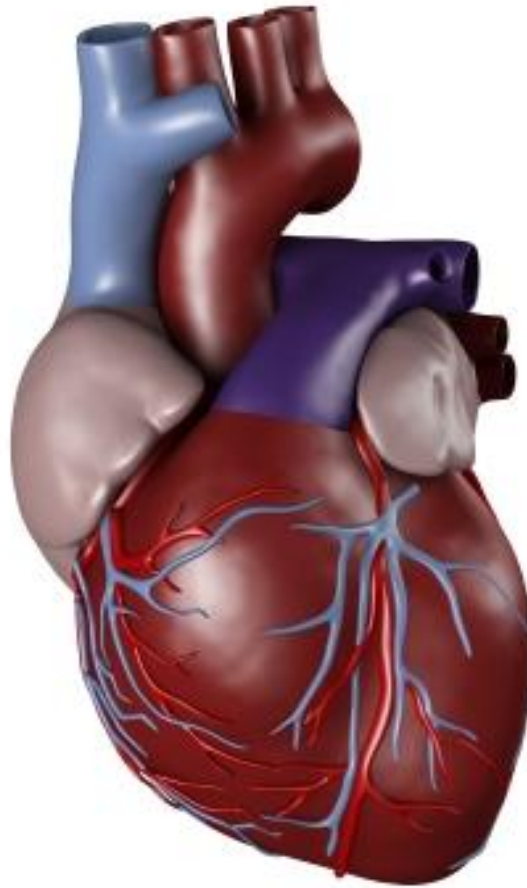
What level is it? plant cell



<http://www.visualanatomy.com/VA/plantcell.jpg>

What level is it?

animal
organ



http://www.biosulf.org/1/images/iStock_heart3699784XSmall.jp

What level is it?

**plant
organ**



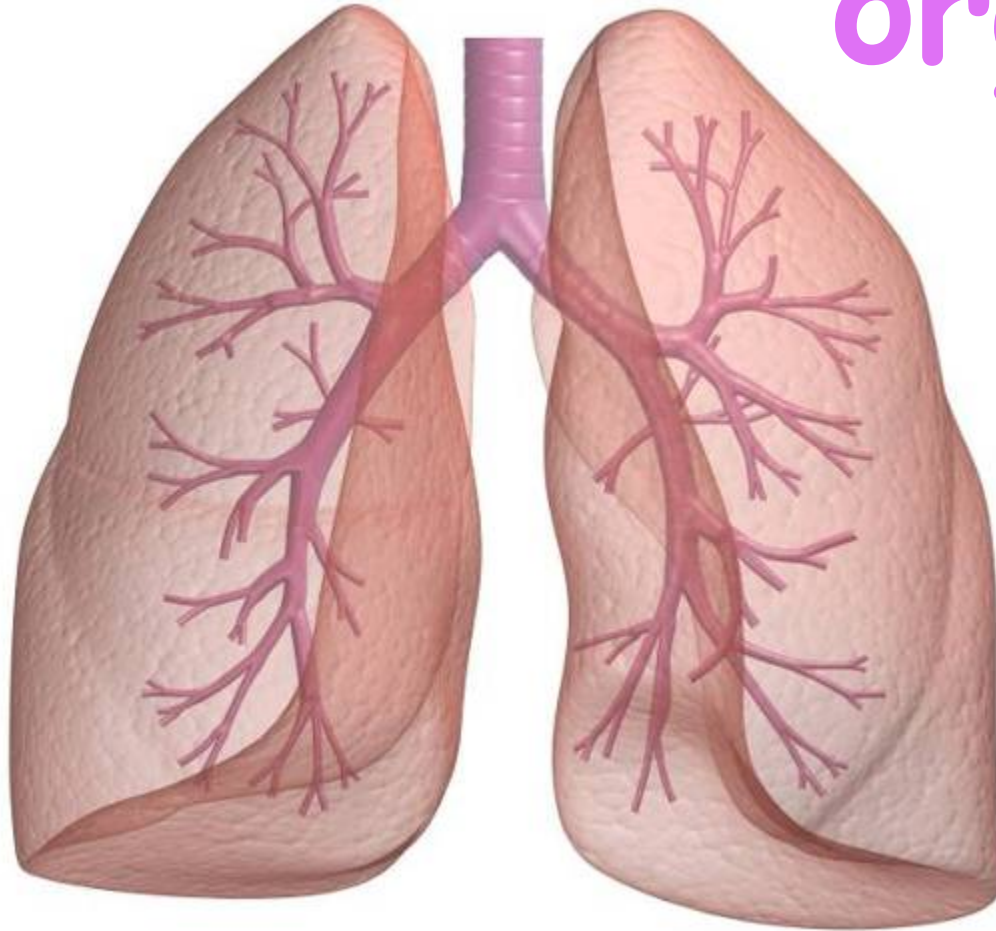
What level is it? **plant
organism**



[http://www.ars.usda.gov/images/docs/7647_7841/2005-X-PlantRoot.jp](http://www.ars.usda.gov/images/docs/7647_7841/2005-X-PlantRoot.jpg)

What level is it?

**animal
organ**



<http://cdn-write.demandstudios.com/upload//3000/700/70/0/313770.jpg>

What level is it?

plant
system



<http://www.arborcentre.co.uk/images/roots.jpg>

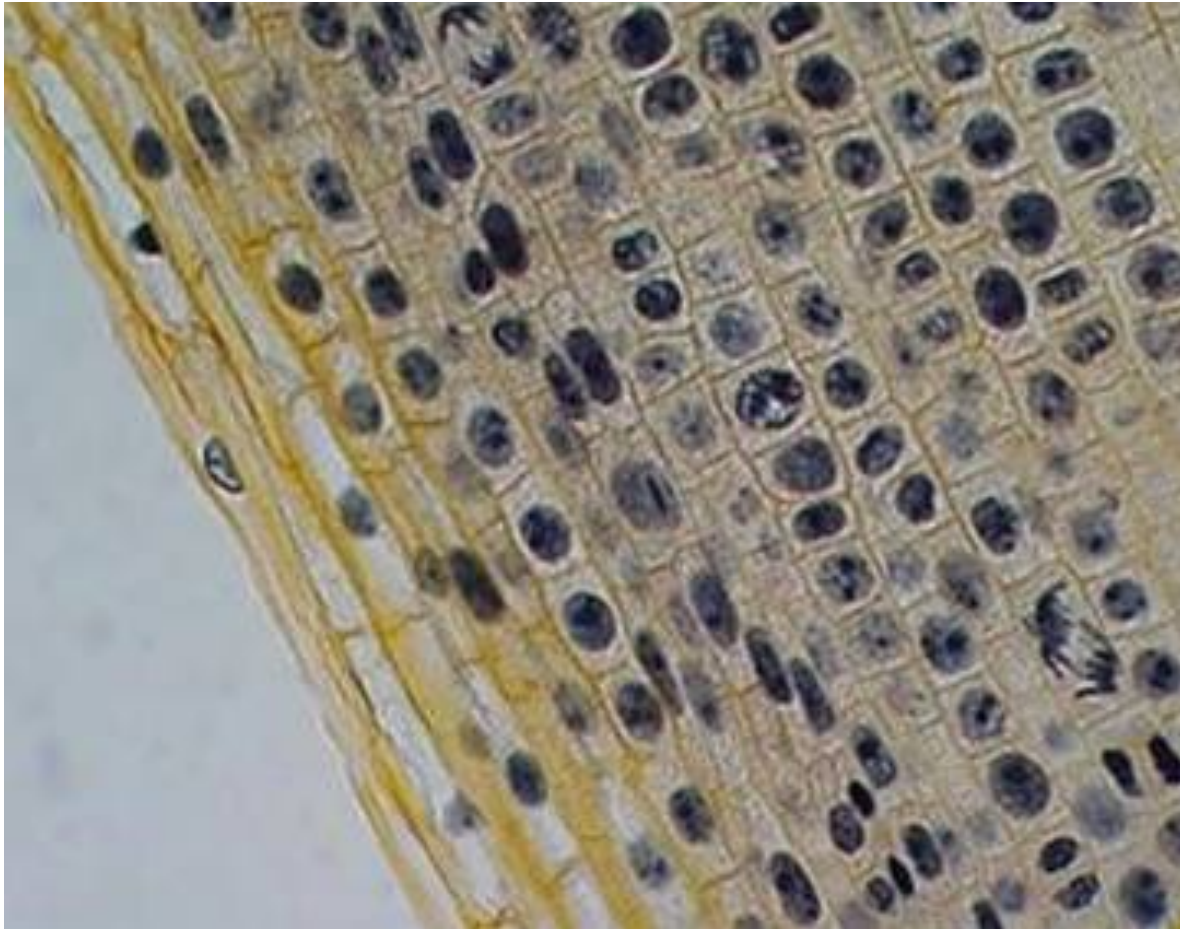
What level is it? **animal
organism**



[http://goldfishbroth.files.wordpress.com/2009/06/greensloth2.jp
g](http://goldfishbroth.files.wordpress.com/2009/06/greensloth2.jpg)

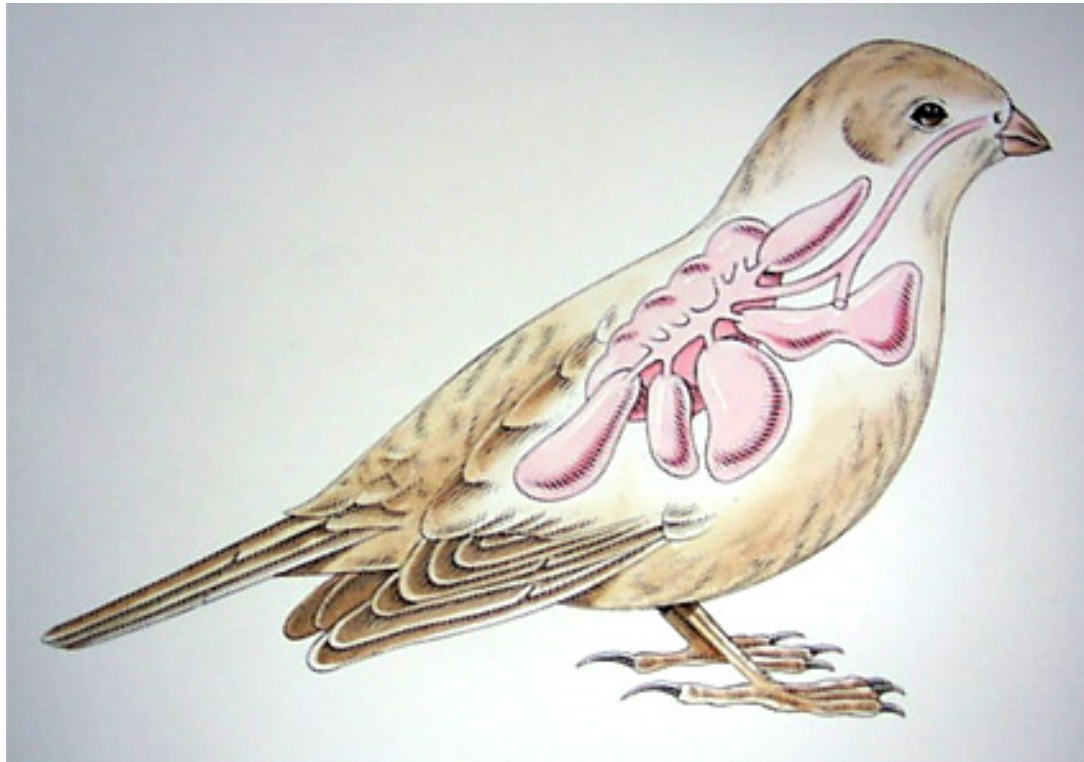
What level is it?

plant
tissue



What level is it?

animal system

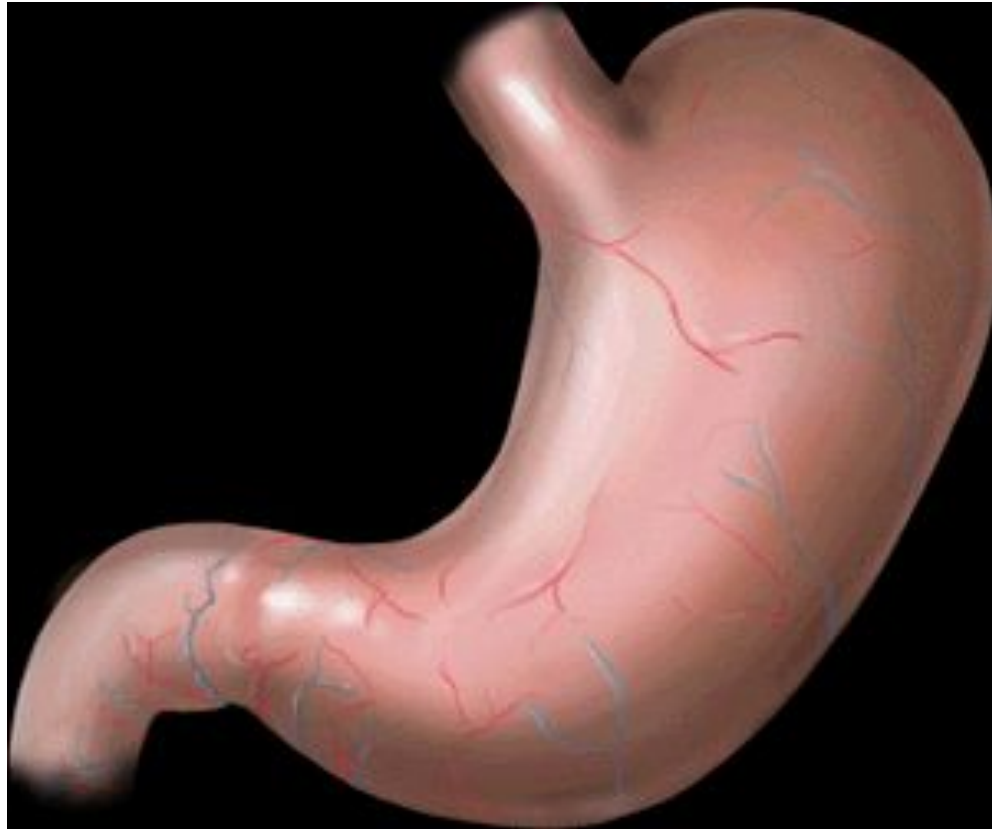


<http://www.dkimages.com/discover/previews/801/910514.JP>

G

What level is it?

animal
organ



http://www.sccollege.edu/SiteCollectionImages/Private/816_stomach.gif

f

What level is it?

plant
organ

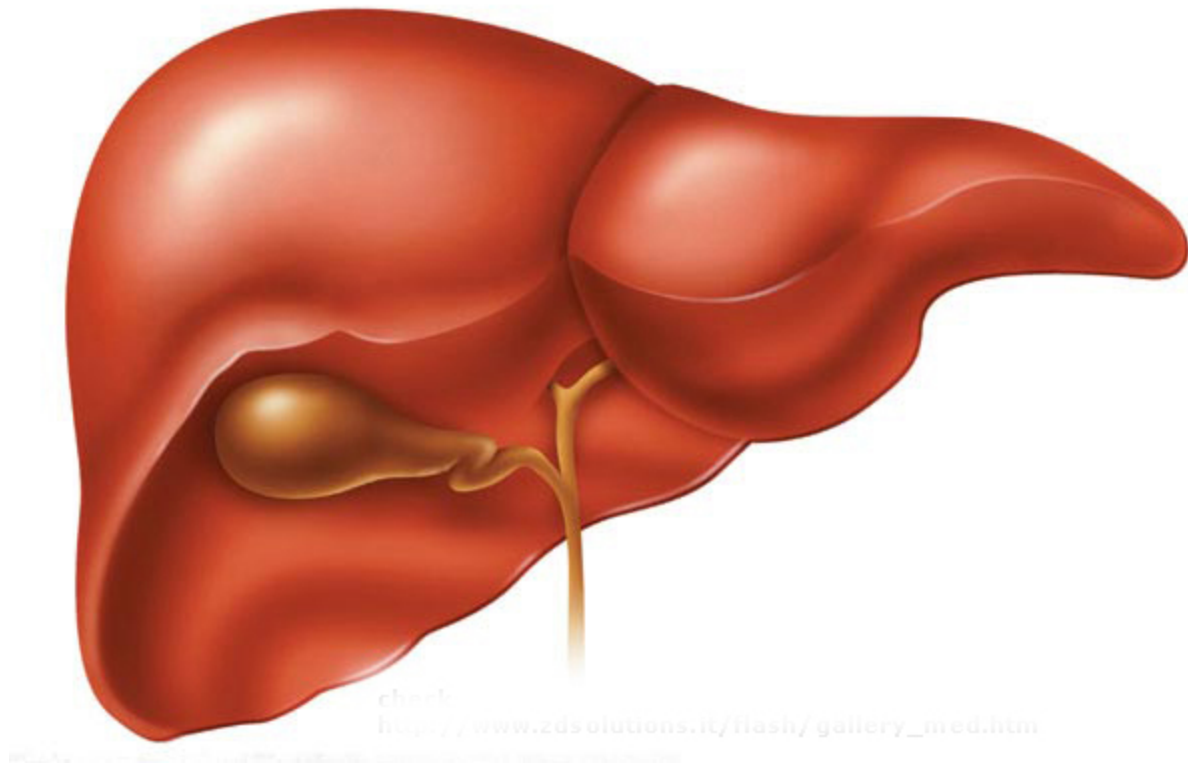


[http://eccstyle.com/blog/wp-content/uploads/2009/07/purple_lotus_flower.jp](http://eccstyle.com/blog/wp-content/uploads/2009/07/purple_lotus_flower.jpg)

g

What level is it?

animal
organ



[http://www.stanford.edu/group/virus/delta/2008/liver.jp](http://www.stanford.edu/group/virus/delta/2008/liver.jpg)

What level is it?

plant
organ

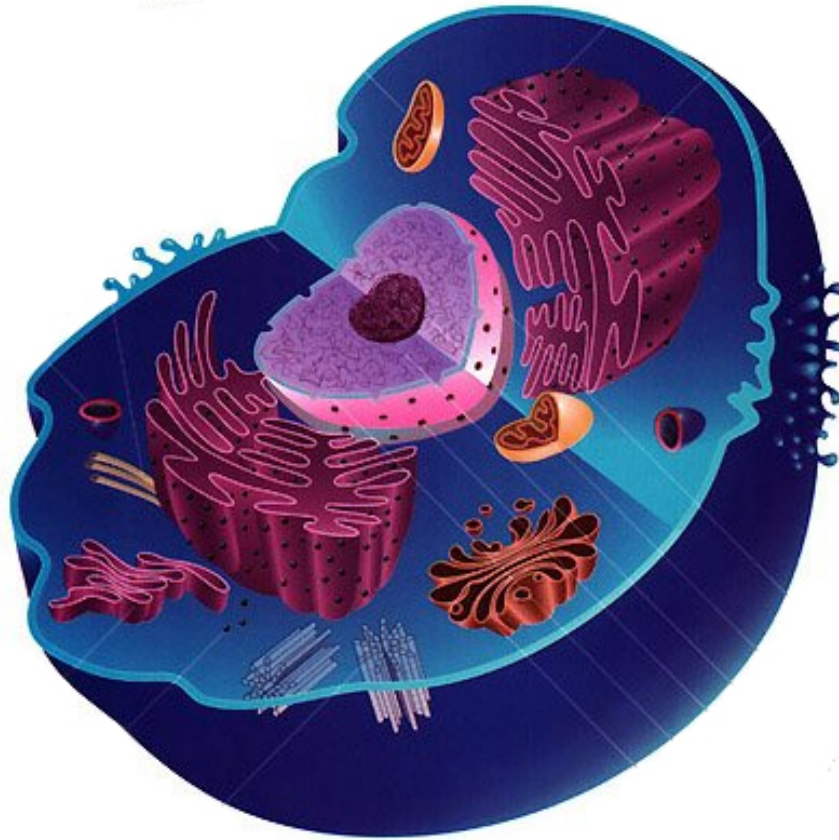


<http://www.gapphotos.com/images/WebPreview/0049/0049655.jp>

g

What level is it?

animal
cell



[http://people.dbq.edu/students/slamey/pictures/AnimalCellColor.jp](http://people.dbq.edu/students/slamey/pictures/AnimalCellColor.jpg)

What level is it?

plant
system



www.wsu.edu/.../delicious/images/spinach.jpg

What level is it?

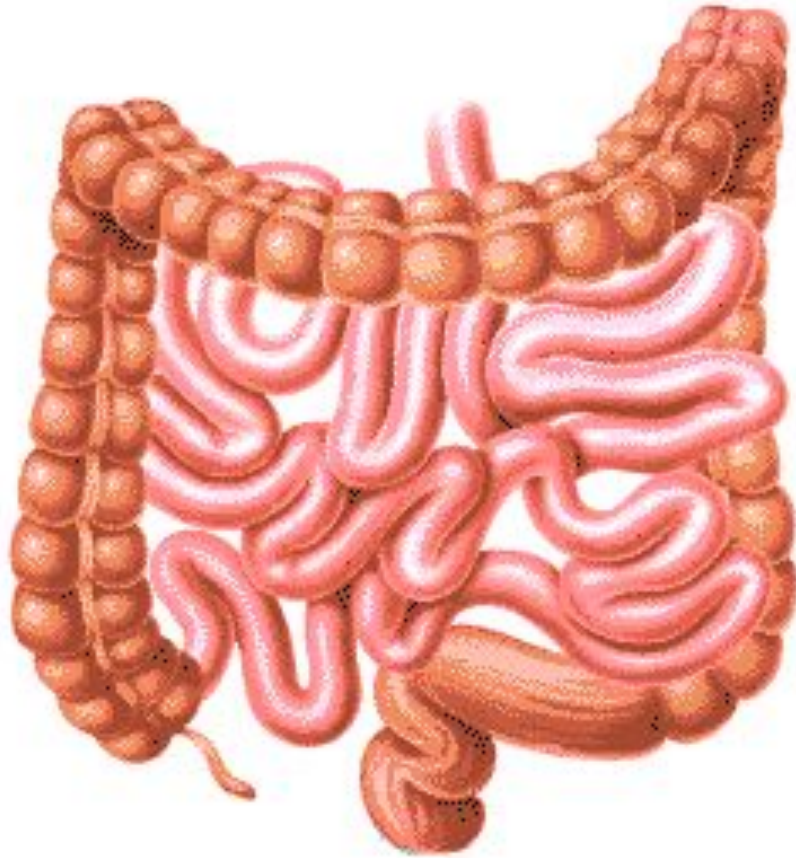
plant
system



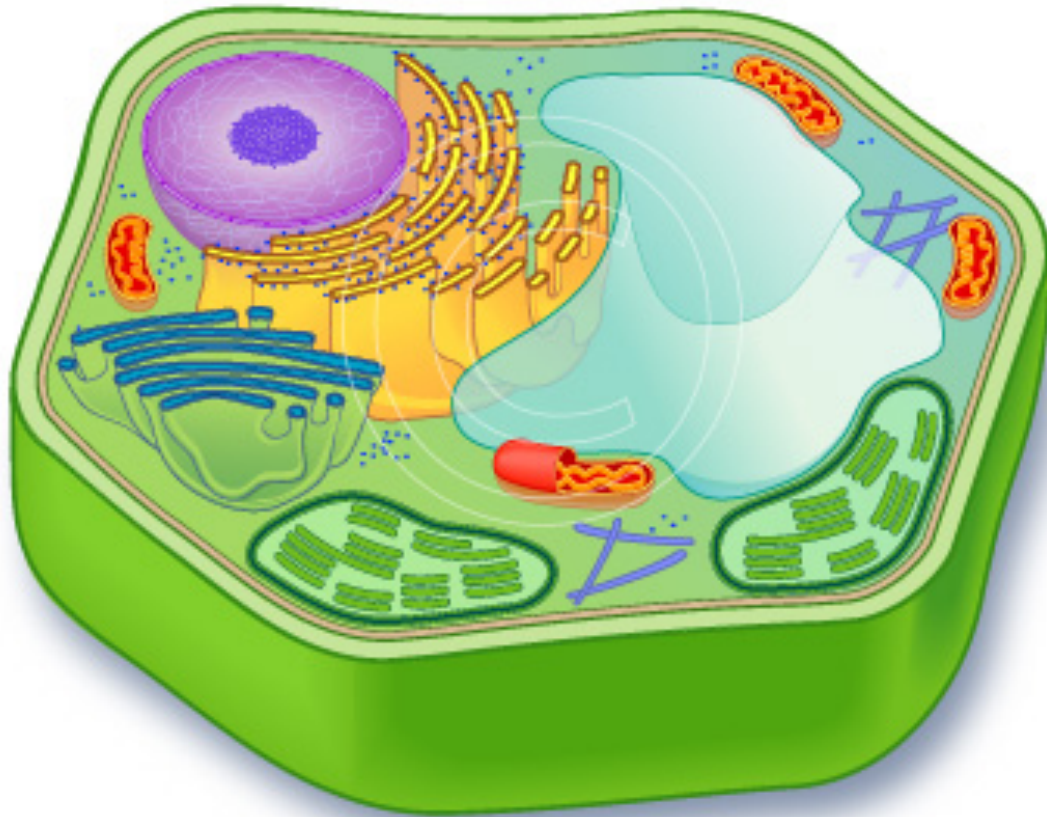
[http://aediconstruction.com/images_solutions/plant_shoot.jp](http://aediconstruction.com/images_solutions/plant_shoot.jpg)
g

What level is it?

animal
organs



What level is it? **plant cell**



[http://syd15.files.wordpress.com/2009/05/plantcell.jp](http://syd15.files.wordpress.com/2009/05/plantcell.jpg)

g

What level is it?

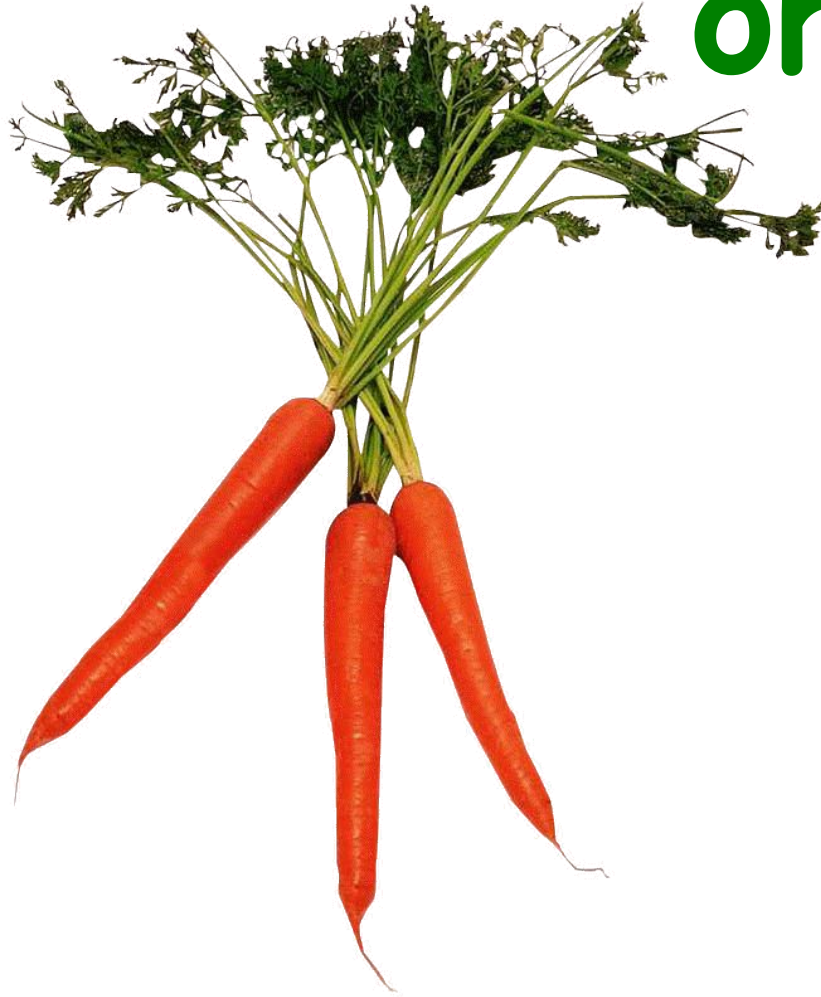
plant
system



<http://aqualandpetsplus.com/Plant,73.jpg>
g

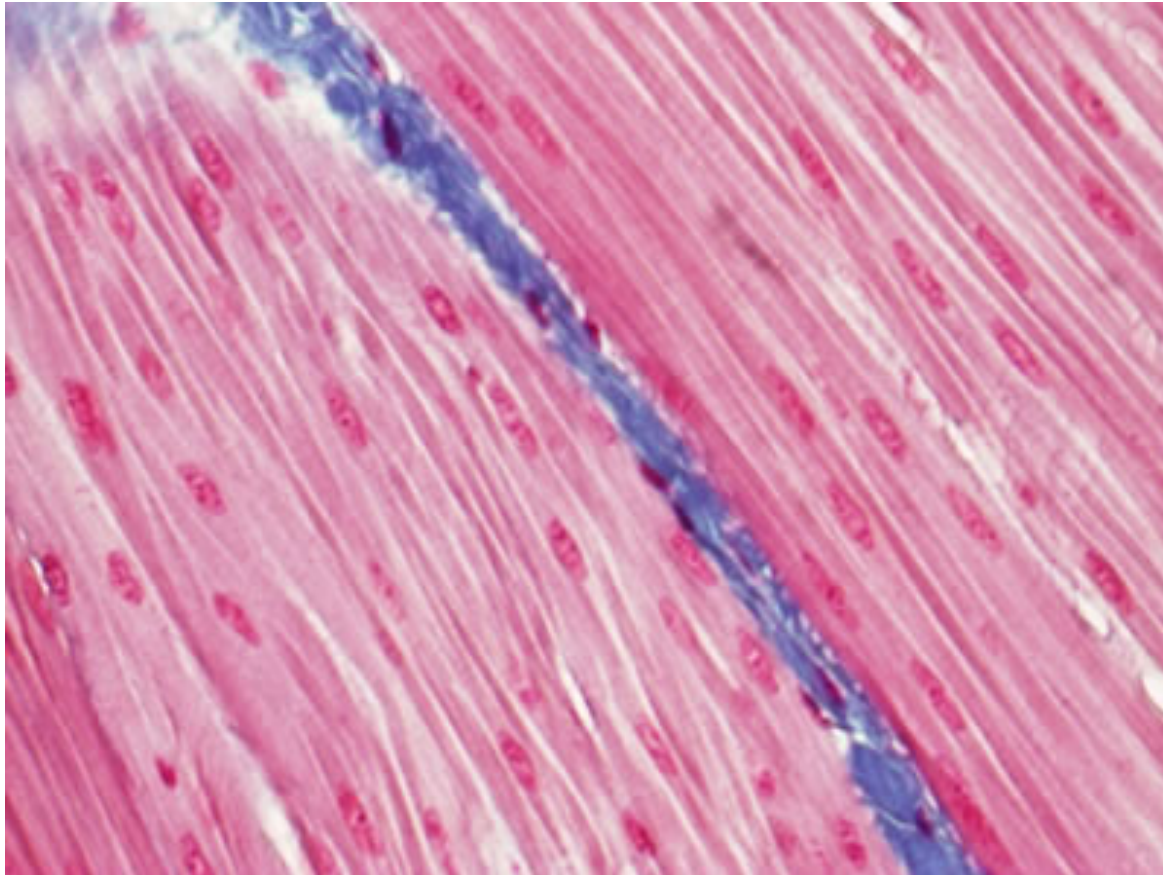
What level is it?

plant
organism



What level is it?

animal
tissue



http://missinglink.ucsf.edu/lm/IDS_101_histo_resource/nerves_muscle.htm

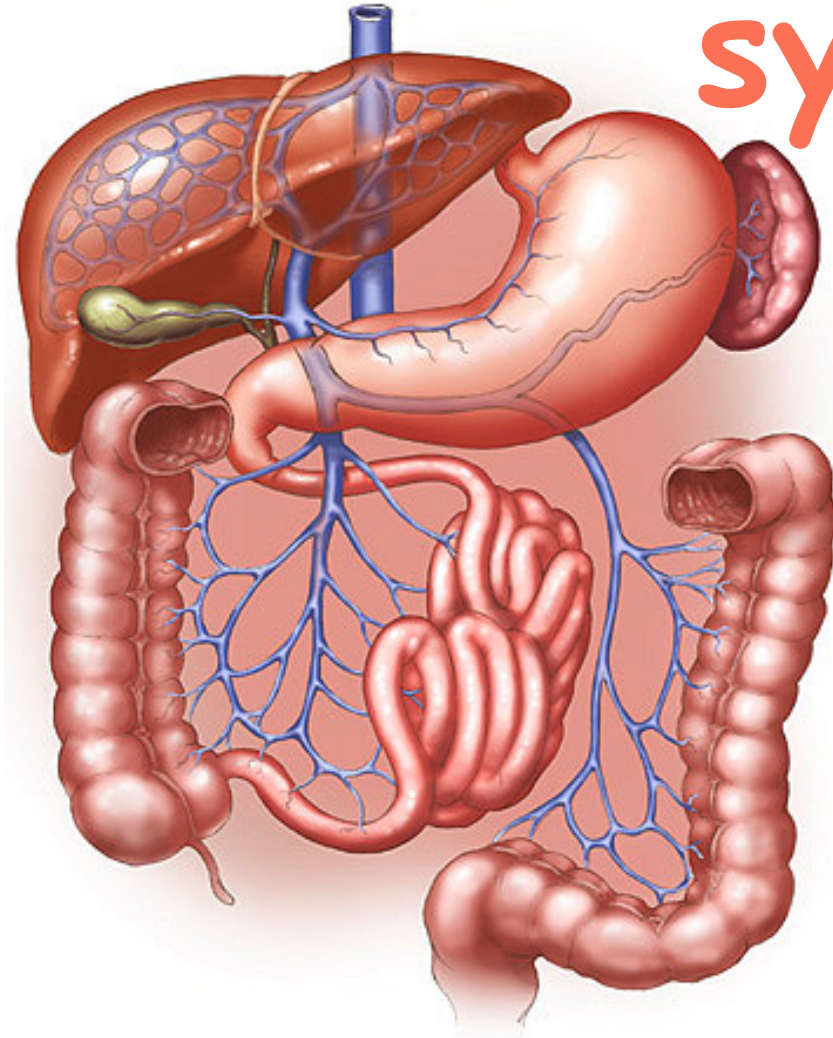
What level is it?

plant
organism



What level is it?

animal
system



What level is it?

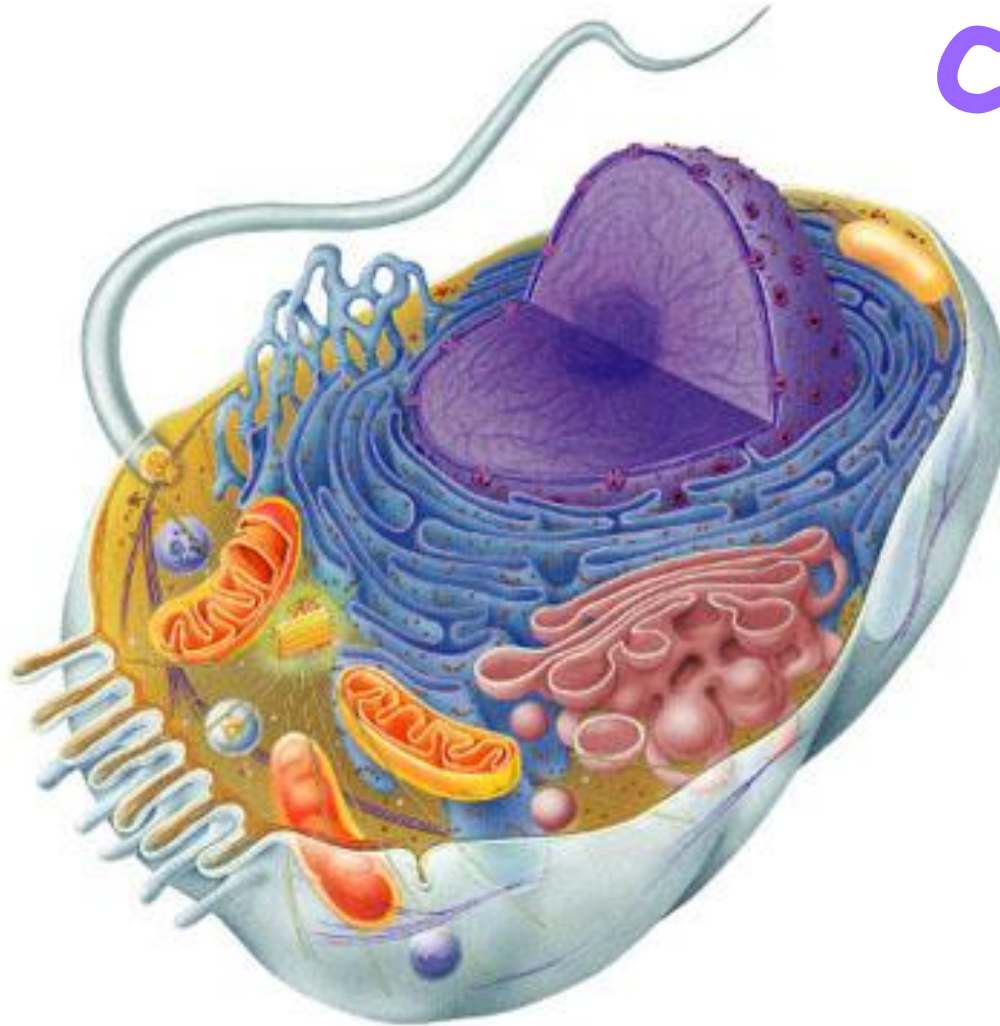
plant
organ



<http://www.foundinthefells.com/monthly/huckleberry%20bud.jpg>

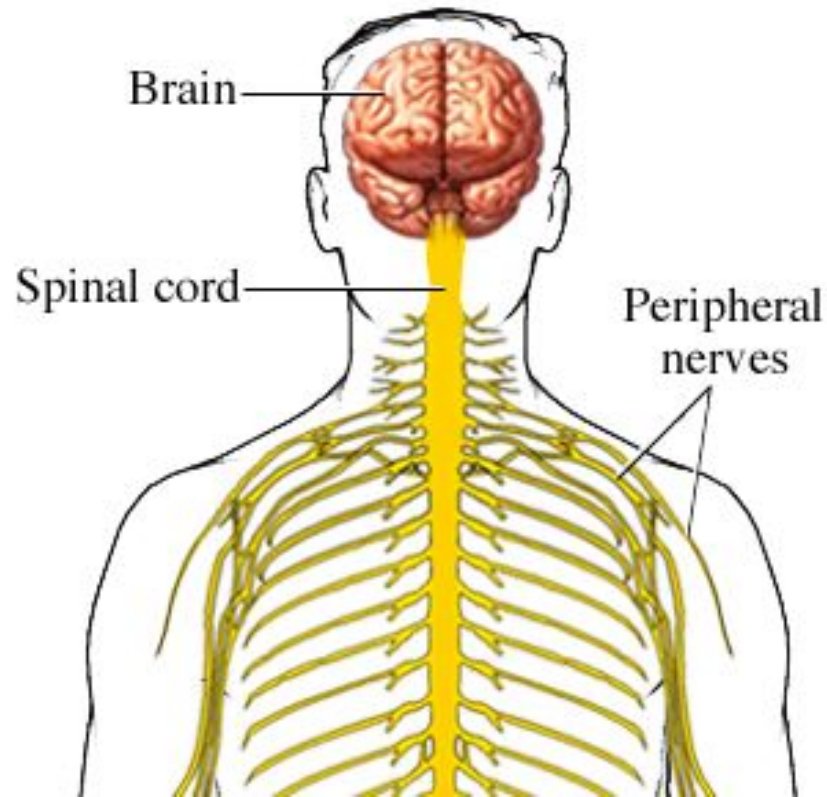
What level is it?

animal
cell



What level is it?

animal system



What level is it?

animal
organ



What level is it?

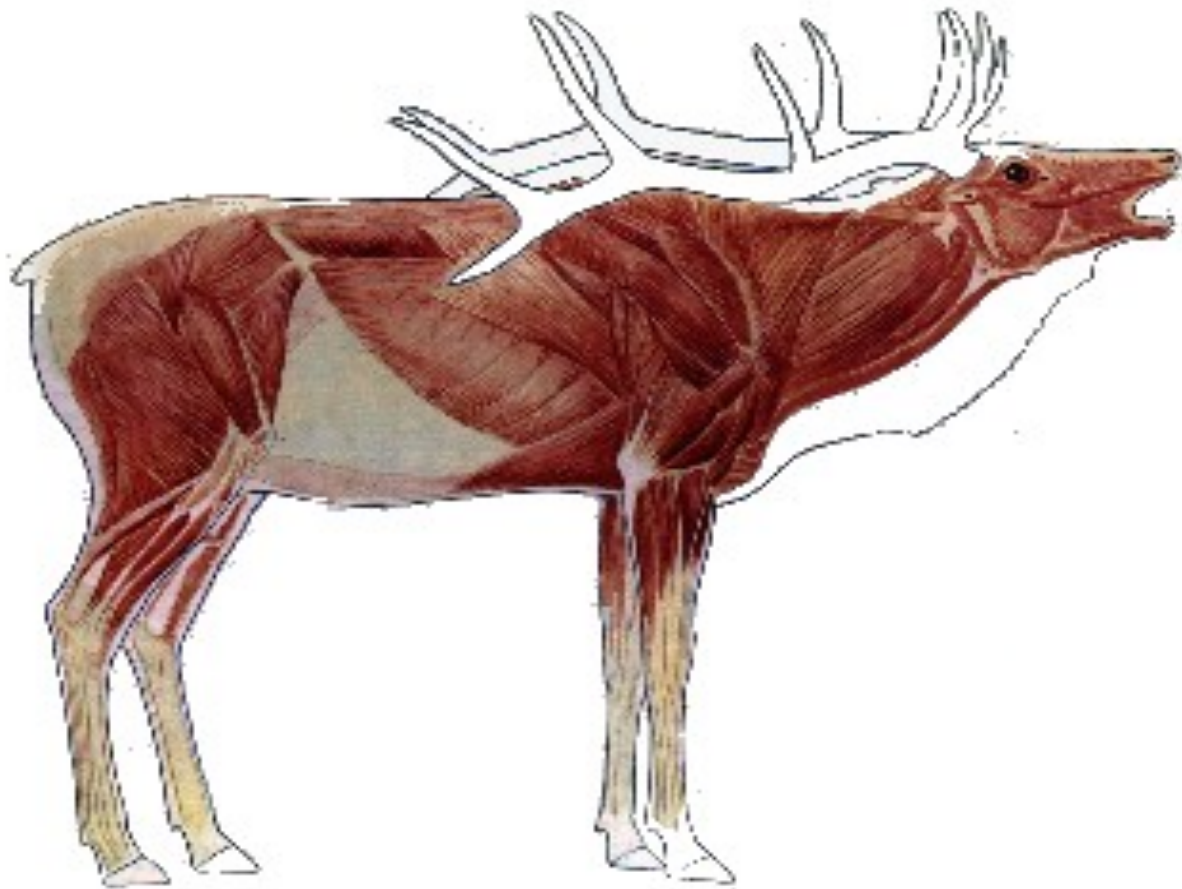
plant
tissue



[http://www.faqs.org/photo-dict/photofiles/list/2090/2735leaf_structure.jp
g](http://www.faqs.org/photo-dict/photofiles/list/2090/2735leaf_structure.jpg)

What level is it?

animal
system



[http://www.bowhunting.net/NAspecies/muscle2.jp](http://www.bowhunting.net/NAspecies/muscle2.jpg)

What level is it? **animal
organism**



[http://trekkingplanetawesome.files.wordpress.com/2009/10/hawksbill-sea-turtle.jp](http://trekkingplanetawesome.files.wordpress.com/2009/10/hawksbill-sea-turtle.jpg)